Korea Technology and Information Promotion Agency for SMEs

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Technology Proposal for 2026 TIPA* International Joint R&D Program

(TIPA: Korea Technology and Information Promotion Agency for SMEs)

The objective of this Technology Proposal is to introduce the institute's proprietary technologies and present potential collaborative areas for Korean SMEs, facilitating joint research and development projects under the 2026 TIPA International Joint R&D Program. This Technology Proposal will be publicly disclosed in conjunction with the official announcement of the 2026 TIPA International Joint R&D Program.

1. Technology Proposer Information

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2. Technology Overview

Title of Proposed Technology	AI-Assisted, Smartphone-Based Dual-Mode Quantitative Urinalysis Platform Using an Au Electrochemical–Optical Integrated LFA Sensor			
Korea's 12 National Strategic Technology Fields	Semiconductor/Display (), Secondary Battery ( ), Advanced Mobility ( ), Next-gen Nuclear Power ( ), Advanced Bio (v), Aerospace/Marine ( ), Hydrogen ( ), Cybersecurity ( ), AI ( ), Next-gen Communication ( ), Advanced Robotics & Manufacturing ( ), Quantum ( )			
Industrial Technology Classification [1]	**Technology Field 1 (**Large Category**)**		**Technology Field 2 (**Large Category**)**	
	Bio·Medical		Electrical·Electronics	
	Medium Category	Small Category	Medium Category	Small Category
	Medical Bio	Reagents/Diagnostic Agents	Measuring Instruments	Measurement Sensors and Components
Concept of Proposed Technology	The proposed technology is a **multi-stage, AI-assisted in vitro diagnostic (IVD) platform** designed to enable **quantitative urinalysis in portable and point-of-care environments**. The platform integrates a **diffusion-free dry-chemistry LFA strip architecture** with an **Au-based electrochemical–optical integrated sensor** and **smartphone-based signal analysis**, forming a unified diagnostic system optimized for both home-based screening and point-of-care testing (POCT). Conventional LFA-based urinalysis systems are primarily limited to qualitative or semi-quantitative analysis due to inherent issues such as reagent diffusion, color bleeding, and cross-interference, particularly when multiple biomarkers are analyzed simultaneously. These structural limitations prevent reliable quantitative signal acquisition and restrict the applicability of LFA technologies in data-driven diagnostic platforms.			

To overcome these limitations, the proposed platform is architected from the outset to support **quantitative signal generation and interpretation**, rather than post-hoc signal estimation.

Technical Architecture: Multi-Stage Diagnostic Platform

Stage 1: Diffusion-Free Dry-Chemistry LFA Strip Architecture

The first stage establishes a diffusion-free dry-chemistry LFA strip designed for stable and reproducible biochemical reactions.
Through optimized multilayer reagent coating structures, controlled membrane properties, and precise spatial configuration of reaction zones, the strip architecture minimizes reagent diffusion and reaction interference.

This structural design enables:

- simultaneous detection of multiple urinary biomarkers,
- stable baseline signal generation, and
- compatibility with both electrochemical and optical signal acquisition.

The strip is optimized for **non-invasive urine sampling**, allowing direct placement on urine collection surfaces (e.g., absorbent pads used in pet urination), thereby supporting home-based screening without additional sample handling.

Stage 2: Au Electrochemical–Optical Dual-Mode Signal Acquisition

In the second stage, the LFA strip is integrated with **Au-based electrochemical sensors** to enable quantitative electrical signal acquisition.
Electrochemical sensing provides sensitivity and numerical precision, while optical signal readout via smartphone imaging offers accessibility and robustness in user environments.

The dual-mode architecture allows complementary signal acquisition:

- electrochemical signals serve as the primary quantitative data source, and
- optical signals function as an auxiliary modality for cross-validation and usability enhancement.

Stage 3: AI-Assisted Smartphone-Based Quantitative Analysis

The final stage integrates smartphone-based data processing and **AI-assisted signal interpretation algorithms**.
Electrical and optical signals are synchronized, normalized, and fused through algorithmic processing to compensate for user variability and environmental factors.

This AI-assisted framework enables:

- conversion of raw sensor signals into reliable quantitative diagnostic outputs,
- consistent performance across diverse user environments, and
- scalable deployment without reliance on dedicated laboratory equipment.

Platform Applicability and Expansion

The proposed platform is initially optimized for **home-based and veterinary urinalysis screening**, where non-invasive sampling and ease of use are critical.

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	At the same time, the system architecture is designed to scale toward **clinical point-of-care diagnostics**, leveraging the same core technologies for broader diagnostic applications. By integrating strip architecture, sensor hardware, and AI-driven analysis into a single platform, the proposed technology establishes a **next-generation quantitative LFA-based diagnostic paradigm**, extending the functional scope of conventional LFA systems. **Conceptual Significance** This concept does not represent a simple combination of existing technologies, but rather a **structurally integrated diagnostic platform** in which: • quantitative capability is embedded at the architectural level, • signal interpretation authority is governed through validated analytical frameworks, and • implementation feasibility is ensured through scalable manufacturing and system integration. Accordingly, the proposed technology provides the conceptual foundation for a globally applicable, quantitatively reliable, and commercially scalable diagnostic platform suitable for international joint R&D.
Rationale for Joint R&D Project	**Structural and Interdisciplinary Complexity** The proposed project requires the structural integration of diffusion-free dry-chemistry LFA strip architecture, Au-based electrochemical–optical dual-mode sensing, and AI-assisted smartphone-based quantitative analysis. These components are not independently modular but must be co-designed to ensure quantitative reliability, analytical consistency, and implementation feasibility. No single organization can independently define analytical validity, implement scalable hardware and manufacturing processes, and govern system-level signal interpretation within this integrated architecture. **Structural Necessity of International Collaboration** International joint R&D is a structural requirement of this project, as it combines the **analytical validation authority and system-level signal interpretation expertise of the overseas research institute** with the **hardware implementation, manufacturing, and system integration capabilities of the Korean SME**. This collaboration enables the realization of a quantitatively reliable diagnostic platform that is simultaneously scientifically validated and commercially implementable, making international joint R&D a necessary execution framework rather than an optional collaboration model.
Key Words	1. Diffusion-Free Dry-Chemistry LFA Platform 2. LFA-Based Quantitative Urinalysis 3. Au Electrochemical–Optical Integrated Biosensor 4. AI-Assisted Smartphone-Based Quantitative Detection 5. Point-of-Care and Home-Based Diagnostics

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3. Research and Development Objectives & Plan

Technology Readiness Level (TRL)[2]	Start of the Project	Level (3)	End of the Project	Level (6)

R&D Objectives

Objective 1. Development of a Diffusion-Free Dry-Chemistry LFA Strip for Quantitative Home-Based Screening

Conventional LFA-based urinalysis strips are primarily designed for qualitative or semi-quantitative interpretation and inherently suffer from reagent diffusion, color bleeding, and cross-interference when multiple biomarkers are analyzed simultaneously. These structural limitations prevent reliable quantitative signal acquisition and restrict the applicability of LFA technologies in data-driven diagnostic platforms.

In this project, a **diffusion-free dry-chemistry LFA strip architecture** will be established as the foundational layer of a quantitative diagnostic platform. This will be achieved through:

- the design of multilayer reagent coating structures that spatially isolate biochemical reactions,
- optimization of membrane materials and pore characteristics to control fluid transport, and
- precise control of reaction order and pad-to-pad spacing to minimize interference.

This strip architecture is not an incremental improvement of existing LFA designs, but a **prerequisite structural condition** for enabling stable and reproducible quantitative signal acquisition using both electrochemical and optical sensing modalities.

Objective 2. Establishment of a Pilot-Scale Manufacturing Process for Multilayer Coating and Reagent Drying

To ensure that the proposed diagnostic platform can transition beyond laboratory-scale demonstrations, the project will establish a **pilot-scale manufacturing process** for LFA strip production. This objective focuses on replacing manual fabrication methods with controlled, repeatable processes suitable for future mass production.

Key development components include:

- automated and reproducible reagent coating processes,
- controlled and uniform drying conditions to stabilize biochemical reactions, and
- defined cutting and packaging workflows aligned with quality control requirements.

Importantly, the pilot-scale process is defined not merely as a production workflow, but as a **data-generation infrastructure**. Manufacturing consistency and process control are essential to ensuring reproducible datasets required for AI-assisted quantitative analysis, directly linking manufacturing quality to analytical performance.

Objective 3. Validation of Au Electrochemical–Optical LFA Sensing Feasibility for Quantitative Diagnostics

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	The core technical challenge of this project is to move beyond visual color interpretation and to enable **quantitative analysis of biochemical reactions on LFA strips** through electrochemical and optical signal conversion. To address this challenge, the project will: • design Au-based Working, Reference, and Counter electrode structures compatible with LFA strip integration, • define stable and reproducible interfaces between strip-based biochemical reactions and electrochemical sensors, and • analyze signal stability, noise characteristics, and environmental variability in both electrochemical and optical modalities. Through **AI-assisted signal interpretation and quantitative validation performed in collaboration with the overseas research partner**, the project will evaluate the feasibility of achieving quantitative diagnostic accuracy within **±5% compared to standard analytical instruments**. This validation will confirm the suitability of the proposed dual-mode sensing architecture as a quantitative diagnostic platform.
R&D Target Specifications	**Diagnostic Scope** **Parameters Measured** The platform targets representative urinary biomarkers including **pH, protein, ketone, bilirubin, and blood**, which collectively cover diverse biochemical reaction mechanisms. These parameters constitute the **minimum required scope** for validating the feasibility of **multi-target quantitative urinalysis** on a strip-based diagnostic platform. **Process Quality** **Performance Metric**: Reagent diffusion **Target Specification**: **0% observable diffusion after contact with urine** This specification is critical to eliminating cross-interference between reaction zones and ensuring data reliability required for quantitative signal interpretation. **Reliability** **Performance Metric**: Signal reproducibility **Target Specification**: **≥95% reproducibility across repeated measurements** High reproducibility ensures minimal signal variation and provides stable, high-quality input data for AI-assisted quantitative analysis. **Shelf Life** **Performance Metric**: Stability of dry-chemistry strip **Target Specification**: **≥12 months at room temperature** This target reflects real-world storage and distribution conditions for home-based and point-of-care diagnostic products.

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	Accuracy **Performance Metric**: Sensor-based quantitative measurement **Target Specification**: **Within ±5% compared to standard analytical instruments** This accuracy target may be further expanded in later validation stages to include correlation metrics (**$R^2 \geq 0.95$**) and reproducibility metrics (**CV ≤ 5%**), establishing analytical equivalence to laboratory-grade reference methods.
R&D Plan	**Reagent & Architecture Development** • Optimization of multilayer reagent coating structures • Evaluation and selection of membrane material properties • Verification of diffusion suppression and reaction isolation • Establishment of a **data-generation architecture** suitable for quantitative analysis **Pilot Process Setup** • Design and implementation of automated coating, drying, and cutting processes • Definition of process-level quality control parameters • Establishment of SOPs and verification of process repeatability • Process stabilization with consideration for future mass production **Sensor Feasibility** • Design of Au-based electrochemical electrode structures (Working / Reference / Counter) • Conversion of strip-based biochemical reactions into stable electrical signals • Verification of electrochemical signal stability for targeted biomarkers • Correlation analysis with optical (smartphone-based) signal outputs **Integrated Validation** • Performance evaluation under user-relevant conditions (lighting, temperature, user variability) • Joint quantitative validation with the overseas research partner • Performance optimization toward global commercialization readiness

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4. R&R (Roles and Responsibilities)

Expected R&R of Research Institute	**Validation (Measurement Principles and Analytical Authority)** Verify the fundamental measurement principles of the proposed diagnostic system using **electrochemical and optical sensing modalities**. Define analytical criteria for distinguishing meaningful signals from noise in strip-based biochemical reactions and provide technical feedback on laboratory-scale performance, including signal stability, sensitivity, and system limitations. **Electrochemical–Optical Signal Interpretation and AI-Based Validation** Provide analytical guidance on the interpretation of **optical and electrochemical signals** generated from LFA-based biochemical reactions. Define validation frameworks for **AI-assisted signal normalization, correlation, and quantitative interpretation**, ensuring that algorithm-derived outputs are scientifically meaningful and analytically defensible. This role focuses on validating **how AI models interpret multi-modal sensor signals**, rather than developing or implementing the algorithms themselves. **System-Level Analytical Consistency and Global Validation Criteria** Establish system-level analytical consistency criteria to ensure reliable quantitative performance across varying user environments and measurement conditions. Define validation metrics and performance benchmarks suitable for international scientific and regulatory evaluation, supporting the analytical credibility of the platform at a global level.
Expected R&R of Korean SME Partner	**Expected R&R of Korean SME Partner (Lead Organization)** The Korean SME acts as the **system implementation and commercialization lead**, responsible for generating reliable quantitative data through hardware design, process control, and manufacturing execution, and for translating validated technology into a market-ready product. **Core Science and Sensor Chip Development** Lead the development of semiconductor process–based sensor chips optimized for strip-based diagnostics. Design and fabricate a new class of LFA-compatible sensor chips that explicitly avoid microfluidic-based architectures, ensuring alignment with data-generation requirements defined in collaboration with the research institute. **Industrial Design and Strip Prototyping** Lead the structural design and prototype development of LFA-based diagnostic strips. Integrate strip architecture with sensor interfaces to ensure mechanical stability and analytical compatibility. Optimize form factors suitable for portable and home-based diagnostic use. **Optimization of Reagent and Membrane Systems** Lead optimization of multilayer reagent coating structures and membrane material properties to control reagent diffusion and ensure reaction uniformity. Ensure consistency of biochemical reactions across strips to support reliable quantitative analysis. **Manufacturing Scale-up and Quality Control**

	Establish pilot-scale manufacturing processes encompassing printing, coating, drying, and cutting. Define and implement quality control (QC) criteria directly linked to analytical performance. Prepare scalable manufacturing workflows suitable for future mass production. **Documentation and Intellectual Property Support** Support intellectual property (IP) generation based on jointly developed technologies. Prepare comprehensive technical documentation covering design, process, and performance. Ensure traceability of technical decisions from R&D through manufacturing and commercialization.

5. Expected Outcomes/Impact

Expected Outcomes and Impact	**Social Impact** This project delivers an **AI-assisted, quantitative home-based screening platform** for companion animal healthcare, enabling earlier detection of health abnormalities and improved access to objective diagnostic indicators. By lowering barriers to routine screening and reducing dependence on subjective visual interpretation, the platform supports **preventive care and timely clinical intervention** in companion animal health through reliable quantitative data. **Technological Impact** The project establishes a **next-generation quantitative in vitro diagnostic (IVD) platform** that integrates a diffusion-free dry-chemistry LFA architecture with **Au-based electrochemical sensing, smartphone-based optical detection, and AI-assisted signal interpretation**. This system-level integration enables robust quantitative signal acquisition under real user environments and advances strip-based diagnostics beyond conventional qualitative or semi-quantitative approaches toward **globally acceptable quantitative diagnostics**. **Industrial Impact** The project strengthens the technological self-reliance of Korean SMEs by securing **core platform capabilities spanning sensor design, strip architecture, manufacturing process control, and quantitative data generation**. Through structured international joint R&D, the project positions Korean SMEs to expand into digital health and preventive healthcare markets with **analytically validated, globally competitive diagnostic platforms**. **Economic Impact** The outcomes of this project support market entry into high-growth veterinary and clinical point-of-care testing (POCT) sectors by establishing a **scalable, globally validated quantitative diagnostic platform**. By reducing technical and regulatory barriers for international expansion, the project creates concrete opportunities for **technology transfer, product commercialization, and sustained participation in global diagnostic markets**.

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6. References

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